

Go Baby Go!

Engineering Accessible Ride-On Vehicles
for Children with Mobility Impairments

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Meet the Team



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Meet the Team



**Claire
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What is Go Baby Go?

Go Baby Go is a program that promotes the use of modified electric ride-on cars as a solution for independent mobility during early childhood for children with disabilities.

Each car is customized to the child's specific physical and cognitive needs, and aims to create a safe, functional, and engaging experience that encourages independent mobility and play.



Ruby's Background

3 year old girl diagnosed with Angelman's syndrome, a neuro-cognitive disorder.

She can independently hold her head up and sit unassisted for extended periods of time.

She is unable to walk unassisted but can pull herself to stand and cruise side-to-side if holding on to something.





Ruby!



Ruby at Meet and Measure Day



Ruby at Trial Day 2

Gantt Chart

Process	Fall Semester										Spring Semester									
	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May										
Intro to Go Baby Go																				
Assigned to Ruby, Meet and Measure																				
1st Presentation																				
Finalize Design																				
Order Materials and Build																				
Test and Finalize Car																				
Demo Day and Final Presentation																				

Methods

Collaborated with child and caregiver to identify user-specific limitations and requirements

Needs Assessment

Selected commercially available ride-on car

Base Vehicle Selection

Designed and refined modifications to optimize safety, usability and comfort

Iterative Design and Prototyping

Evaluated user-interaction through multiple trial days with Ruby and her family.

Validation Testing

Adapted seating and support structures, and rewired control system to incorporate accessible input devices

Mechanical, Electrical and Control Modifications

General Materials



PVC Tubing and Fittings

Lightweight, strong, and easy to cut or shape. Used for structural supports, mounting points, or custom steering adaptations.

Foam Blocks/Padding

Used to create back and side supports, improve seating comfort, and provide trunk stabilization. Easy to contour and safe for pediatric use.

3D Printed PLA Components

Allows for precise, custom-shaped parts such as brackets, handles, or enclosures. PLA is rigid, easy to print, and ideal for low-load adaptive components.

Wiring and Connectors

Used to rewire the original foot pedal accelerator to hand-activated push button on steering wheel and connect the emergency stop.

Safety and Control Components

Master Shut-Off Switch

Provides caregivers with immediate control of the vehicle's power. Allows the car to be disabled instantly in case of unsafe driving or unexpected behavior.



Speed Limiter

Restricts the top speed of the vehicle to ensure safe operation. Maintains an appropriate and controlled pace for a three-year-old with mobility and balance challenges.

Ruby's Vehicle Modifications

Hand-Activated Steering Wheel Accelerator System

Foot pedal accelerator rewired to a push-button mounted on the steering wheel to enable upper-extremity control.

Custom PVC Postural Support Frame

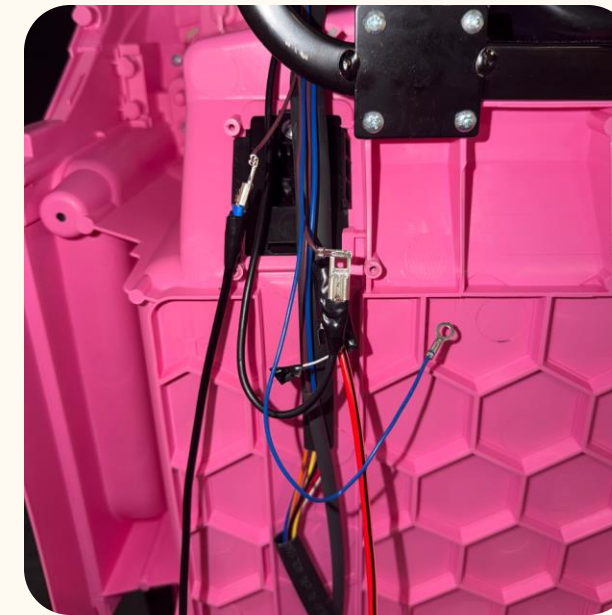
Rigid rear support structure constructed from PVC to enhance trunk stability and alignment.

Adaptive Swim Board Backrest

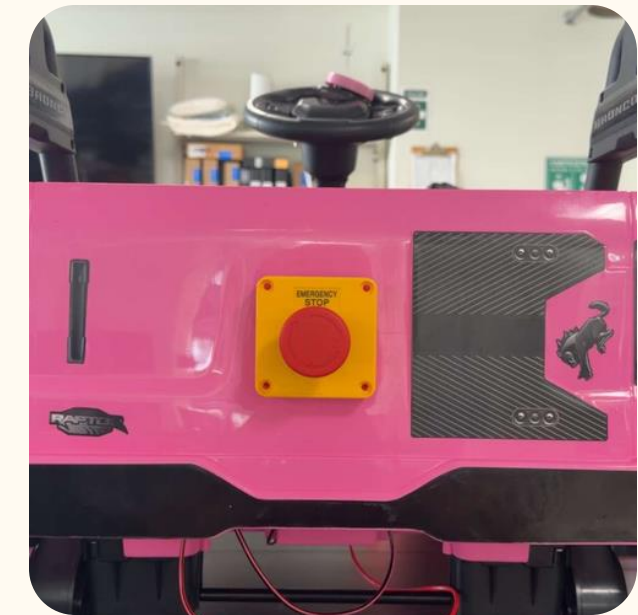
Contoured back support integrated to improve comfort and maintain upright posture.

Five-Point Safety Harness System

Secures the user at the shoulders, hips, and pelvis to maximize safety and positioning.



Completed
rewiring



Rear-mounted emergency
stop



Support frame, backrest, and
safety harness



Hand-activated steering wheel
accelerator

Ruby's Vehicle Modifications

Electronic Speed Limiting System

Reduces maximum vehicle speed to ensure safe and controlled operation.

Integrated Steering-Assist Activation Bar

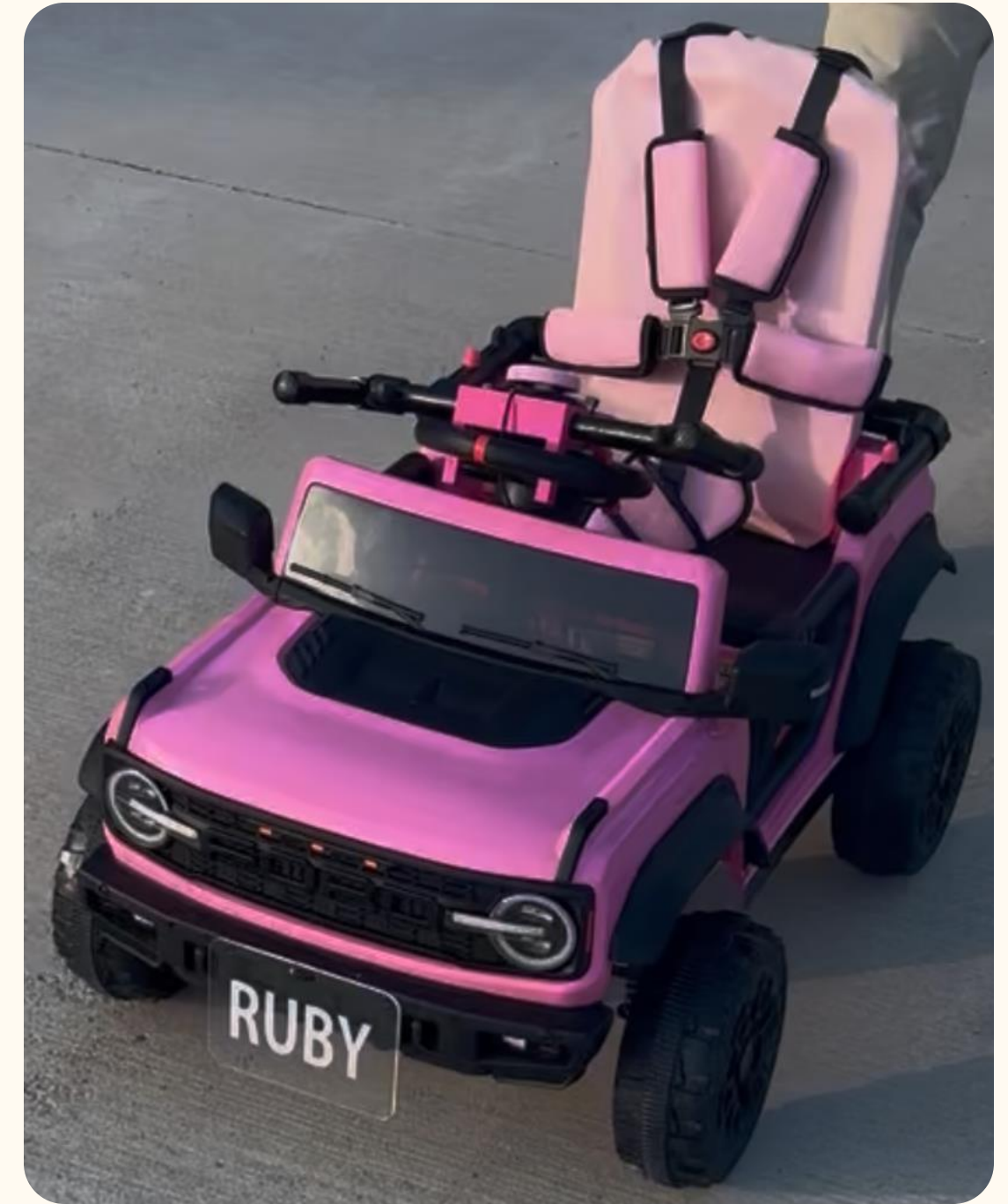
Support bar positioned over the accelerator button to facilitate simultaneous steering and activation.

Rear-Mounted Emergency Stop System

Easily accessible kill switch located on the back of the vehicle to allow immediate intervention by a caregiver.

Parental Remote-Control Override System

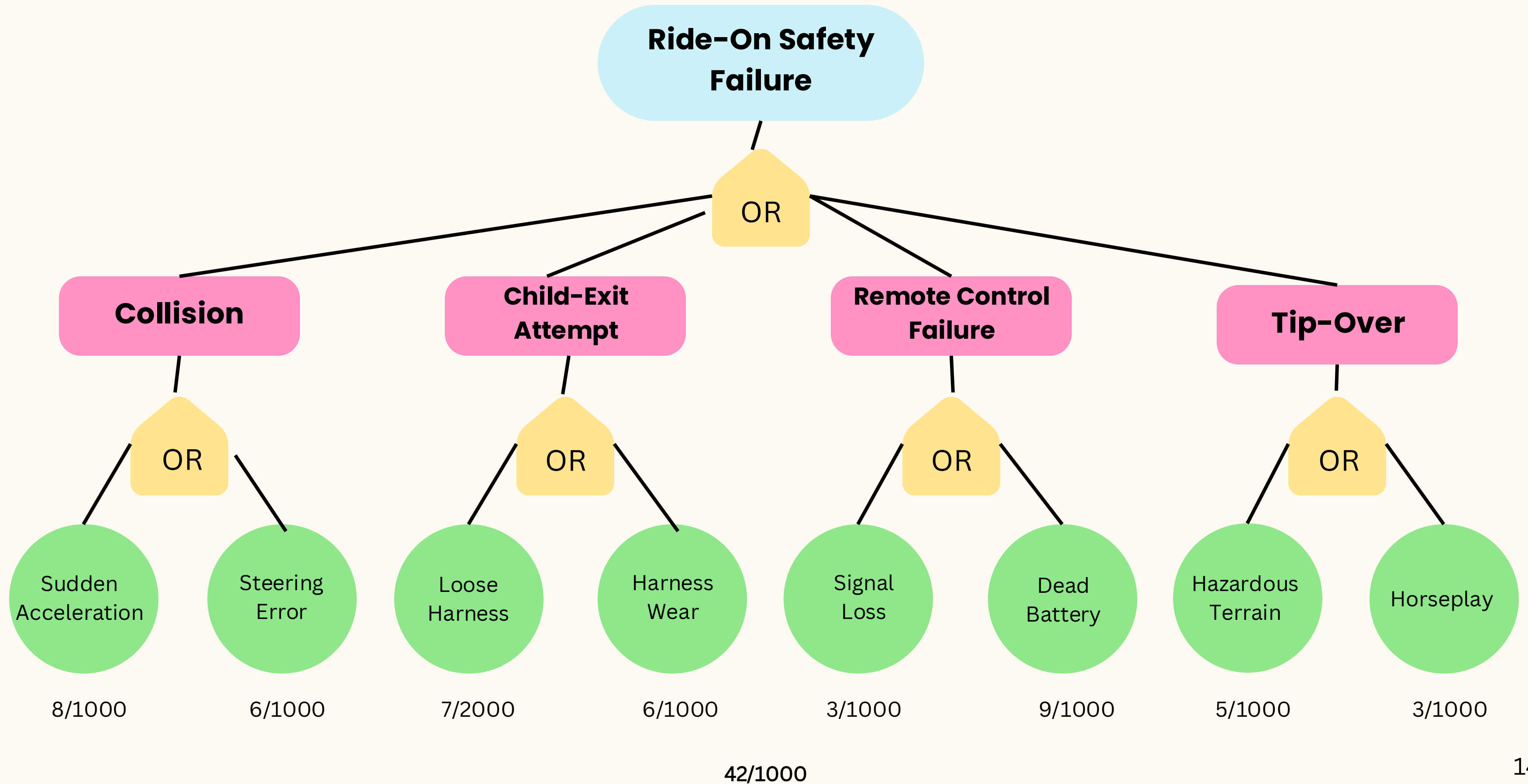
Wireless remote enables caregiver control of the vehicle when the emergency stop is engaged, ensuring safe supervision and assisted mobility.



The completed vehicle at Expo Day

Risk Analysis	Event/Gate	Probability	Event Short Name	Explanation
	Top	42/1000	Ride-On Safety Failure	System-level failure causes ride-on car to be unsafe for child to ride.
	G1	13/1000	Collision	Child or car collides with obstacle due to limited attention or delayed reaction
	G2	8/1000	Tip-over	Car tips due to sharp turn, uneven terrain or weight shift
	G3	9/1000	Child Exit Attempt	Child attempts to stand, climb, or lean out of moving vehicle
	G4	12/1000	Remote Control Failure	Parent remote override disconnects or fails to respond
	E1	8/1000	Sudden Acceleration	Motor unexpectedly accelerates due to button press or component fault
	E2	6/1000	Steering Error	Child over-steers or steers unpredictly
	E3	3/1000	Remote Signal Loss	Bluetooth/RC signal temporarily fails
	E4	9/1000	Remote Battery Dead	Remote power dies mid-operation
	E5	7/1000	Harness Improperly Buckled	Child improperly secured in harness
	E6	5/1000	Harness Wear/Tear	Harness component degrades or detaches
	E7	5/1000	Hazardous Terrain	Slope, hole, curb, rocks, unstable surface, etc
	E8	3/1000	Horseplay	Child hanging out of vehicle or doing task near/in vehicle that can cause balance of car to change drastically.
				13

Fault Tree Analysis



Key Discussions



Commercially available vehicles can undergo low-cost modifications to be transformed into functional assistive mobility devices, effectively improving accessibility and independent mobility for pediatric users.

The design adaptations improved comfort, safety, ease of use, and increased engagement with the environment.





Future Directions

As our design is tailored to a specific user, broader applicability may require long-term testing and additional customization.

Longitudinal data and developmental impact studies may help provide this information.





Ruby at Distribution Day!



Thank
you!